Enterprise AI Framework (EAIF) v1.0

What EAIF Is

> **EAIF is a technology-independent architecture framework for designing AI-enabled business capabilities and shared AI platforms that remain useful as AI technologies change.**

Four Enduring Questions

CONTROL	EXECUTE	ASSURE	EVOLVE
Should this happen, and under what conditions?	How is useful work performed?	Can we trust what happens?	What evidence should make it better?

Intended Audience

Enterprise, AI, Solution, and Platform Architects; Staff and Principal Engineers; AI platform teams; and engineering leaders designing enterprise AI capabilities.

Scope

EAIF defines architectural responsibilities and boundaries, not implementation technologies. Models, retrieval techniques, agents, workflows, tools, protocols, and platforms fit beneath those responsibilities and may change without requiring the architecture itself to change.

WHAT EAIF IS	WHAT EAIF IS NOT
• Enterprise AI architecture framework • Technology-independent North Star • Common architectural language • Guide for separating reusable mechanisms from domain meaning	• AI risk-management or governance framework • Delivery methodology or maturity model • Reference implementation or vendor architecture • Certification or conformance standard

EAIF operates within applicable enterprise governance, risk, legal, regulatory, security, and domain requirements. It does not replace the processes that establish them.

> **EAIF provides a durable architecture for enterprise AI without making today's AI technologies the architecture.**

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Design Philosophy + Core Principles

Design Philosophy

TIMELESS	SIMPLE	BORING	LEAN
Separate enduring responsibilities from temporary mechanisms.	Introduce only the concepts required to make useful architecture decisions.	Reuse established enterprise engineering where it already solves the problem.	Make every shared capability, abstraction, and autonomous boundary earn its place.

Core Principles

1. Business Value First

Start with the business capability and intended outcome, not the AI technology. Use AI when it materially improves the result. Prefer simpler mechanisms when they are sufficient.

2. Centralize Mechanisms, Decentralize Meaning

Share reusable mechanisms where consistency, trust, cost, or operability improve. Keep business semantics, authoritative knowledge, domain rules, and business correctness with the domains that own them.

3. Trust Through Evidence

Trust the complete capability through evidence of how it behaves for its intended purpose and consequence.

> **Build for demonstrated need. Preserve the ability to change.**

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EAIF Core Architecture

4 domains · 10 capabilities

EAIF consists of four architectural domains and ten capabilities. Each domain answers an enduring architecture question.

CONTROL	**Should this happen, and under what conditions?**	**Identity · Purpose · Policy**
EXECUTE	**How is useful work performed?**	**Reason · Access · Act**
ASSURE	**Can we trust what happens?**	**Protect · Evaluate · Observe**
EVOLVE	**What evidence should make it better?**	**Improve**

What Each Domain Does

Control: Establish the context and boundaries within which execution may operate.

Execute: Perform the business capability through reasoning, authorized information access, and permitted operations.

Assure: Protect execution, evaluate whether outcomes are fit for purpose, and preserve sufficient operational evidence.

Evolve: Use evidence to improve, simplify, constrain, replace, or remove mechanisms when appropriate.

> **EAIF defines architectural responsibilities. It does not prescribe how the system must be built.**

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Understanding the Core

EAIF describes responsibilities, not a mandatory system topology.

NOT A PIPELINE

Control, Execute, Assure, and Evolve are not four sequential processing stages. Execution may revisit concerns when authority, purpose, consequence, or proposed action changes.

NOT SERVICES

The ten capabilities do not imply ten services, teams, deployment units, platform components, or architectural planes.

USE WHAT THE WORKLOAD NEEDS

EAIF provides a common vocabulary, not a mandatory feature checklist. A read-only capability may use Reason and Access without Act.

RESPONSIBILITIES STAY STABLE

Implementation mechanisms can change while the architectural responsibility remains the same.

Selected Responsibilities and Example Mechanisms

Responsibility	Example mechanisms
Reason	Models, rules, deterministic logic, planning, agents
Access	APIs, search, retrieval, databases, semantic layers
Act	Tools, workflows, enterprise APIs, systems of record
Evaluate	Tests, domain datasets, validation, human judgment
Observe	Logs, metrics, traces, audit and operational evidence

> **Use existing enterprise mechanisms where they already satisfy the responsibility.**

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Control: Identity, Purpose, Policy

> **Control establishes the conditions within which execution is permitted to operate.**

Identity

Who initiated the work, who or what is currently acting, and under whose authority?

Identity preserves the relationship between initiator, current actor, and authority. For delegated execution, authority may narrow but does not expand.

Purpose

What business outcome or task is this execution serving?

Purpose defines why execution is occurring and provides context for the applicable boundary. A delegated task may have a narrower purpose than the original request.

Policy

What rules, permissions, constraints, and conditions apply?

Policy may represent permitted information and actions, delegation boundaries, resource limits, read-only or state-changing behavior, approval requirements, and prohibited operations.

Delegated Execution



> **Delegation does not expand authority.**

Dynamic Execution

Control is not necessarily established once. When execution materially changes purpose, authority, consequence, or proposed action, applicable boundaries should be re-evaluated.

Policy consumes applicable enterprise, domain, security, risk, legal, and regulatory requirements. EAIF does not require Policy to replace the systems that establish or enforce those requirements.

> **Greater reasoning capability does not imply greater authority.**

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Execute: Reason, Access, Act

> **Execute performs the business capability within the boundaries established by Control.**

REASON	Determine what should be done and how intelligent processing should contribute.	Models · rules · deterministic logic · classification · planning · agents
ACCESS	Obtain the authorized information and context required for execution.	Databases · search · retrieval · RAG · semantic layers · APIs
ACT	Perform permitted operations through enterprise capabilities, tools, workflows, or systems.	Tools · workflows · enterprise APIs · systems of record

Boundaries

- Models and agents are mechanisms beneath Reason, not EAIF domains.
- RAG is one Access pattern. Access obtains information; it does not establish authority or own domain meaning.
- Act causes an operation to occur. Consequential operations should continue through authoritative enterprise systems.
- Orchestration coordinates execution across Reason, Access, Act, and applicable Control and Assure boundaries.

> **Reason what is needed. Access what is permitted. Act only where authorized.**

Not every intelligent capability needs Act. Read-only analytics, summarization, classification, and knowledge capabilities may operate through Reason and Access alone.

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Assure: Protect, Evaluate, Observe

> **Assure establishes appropriate trust in intelligent execution.**

PROTECT	Keep execution within the boundaries under which it was permitted.	Authorization · data protection · tool restrictions · containment · validation · approval
EVALUATE	Determine whether behavior and outcomes are sufficiently fit for purpose.	Business correctness · quality · policy adherence · reliability · domain criteria
OBSERVE	Preserve sufficient evidence to operate, understand, and account for material execution.	Actors · authority · delegation · actions · approvals · failures · outcomes

Assurance Rules

- Higher consequence justifies stronger assurance.
- Evaluate end to end for confidence. Evaluate components for diagnosis.
- Observe enough to reconstruct material responsibility, not every intermediate interaction.
- Protect, Evaluate, and Observe may apply before, during, and after execution.
- When Policy requires approval, Assure establishes that the condition is satisfied before Act commits the governed operation.

> **Observation should support operation and accountability without indiscriminate retention.**

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Evolve: Improve

Evolve uses evidence to change the capability when change creates meaningful value.

Improve

Use production, evaluation, operational, and business evidence to determine what should change. Evidence may include business outcomes, evaluation results, incidents, user corrections, failures, latency, reliability, cost, recurring execution paths, and changing domain requirements.

Improve the capability, not merely the AI.

ADD Introduce a mechanism when evidence demonstrates need.	**CHANGE** Modify an existing mechanism to better fit the requirement.	**SIMPLIFY** Remove unnecessary architectural or execution complexity.
CONSTRAIN Reduce freedom, scope, access, autonomy, or resource usage.	**REPLACE** Substitute another intelligent or deterministic mechanism.	**REMOVE** Retire AI or the capability when value no longer justifies it.

Evolution Can Move in Both Directions



Evidence may justify greater autonomy or greater determinism. More AI is not inherently more mature architecture.

Let evidence determine what evolves. Let business value determine what remains.

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Platform vs Domain

> **Centralize Mechanisms. Decentralize Meaning.**

Enterprise AI platforms create value by sharing mechanisms that should not be rebuilt independently by every domain. Centralization becomes harmful when the platform begins owning business meaning.

Mechanism vs Meaning

Area	Shared mechanism	Domain meaning
Identity & Policy	Integration, enforcement, common controls	Domain-specific permissions and constraints
Reasoning	Approved access, lifecycle, execution	Required business quality and appropriate use
Knowledge & Data	Retrieval, query, access, validation	Authoritative sources and semantics
Evaluation	Execution, regression, comparison, reporting	Scenarios, expected outcomes, correctness
Observability	Common telemetry and evidence	Domain interpretation and business impact
Actions	Integration and execution mechanisms	Business rules and authoritative state

Configure → Extend → Specialize

CONFIGURE	EXTEND	SPECIALIZE
Use supported configuration of the shared mechanism.	Improve the shared mechanism when the need is reusable across domains.	Keep the capability domain-specific when reuse is low or generalization would absorb business meaning.

> **The platform provides reusable mechanisms. The domain owns meaning.**

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Workflows and Agents

EAIF treats workflows and agents as execution patterns beneath Execute.

> **Does the execution path itself need to be decided at runtime?**

Deterministic Workflow vs Autonomous Agent

	Deterministic workflow	Autonomous agent
Path	Primarily predefined	Partly runtime-decided
Reason	Within known steps	May select and replan steps
Access	Usually predetermined	May select permitted sources
Act	Explicitly orchestrated	May select permitted actions
Control and Assure	Required	Required

Decision Rule

Use a workflow when the required path is known. Consider an agent when runtime reasoning over the execution path materially improves the outcome.

- Do not use intelligence to rediscover a process the system already knows.
- Autonomy remains bounded by Control and Assure.
- Production evidence may justify Workflow → Agent or Agent → Workflow.

> **Use workflows when the path is known. Consider agents when the path itself requires runtime reasoning.**

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Delegation and Human Approval

Delegated and consequential execution use the same EAIF Core.

Delegation



Delegation changes who performs the work. It does not create broader authority. Before adding another autonomous participant, ask whether independent reasoning at that boundary materially improves the outcome.

> **Delegation does not expand authority.**

Simplification Bias



This is not a maturity model.

> **Agent consensus is not governance.**

REASON	PROPOSE	VALIDATE	APPROVE	COMMIT
Determine	Make explicit	Check rules	Satisfy authority	Perform action

Policy determines whether approval is required. Assure establishes that required validation and approval occurred. Act performs the permitted operation through the authoritative system.

> **Reasoning may propose. Authority determines whether execution may commit.**

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EAIF and NIST AI RMF

> **NIST AI RMF provides a framework for managing AI risk.**
> **EAIF structures enterprise AI architecture.**

The relationship is many-to-many. This table lists the strongest NIST AI RMF relationship for each EAIF domain.

Crosswalk

EAIF domain	Strongest NIST relationship
Control	Govern + Map
Execute	Map
Assure	Measure + Manage
Evolve	Manage + Measure

Govern is cross-cutting and should not be treated simply as the NIST equivalent of Control.

Genuine Scope Differences

NIST IS BROADER IN

- Organizational governance and accountability
- Formal risk identification and treatment
- Affected-party and societal impacts
- Third-party risk management
- Formal feedback, appeal, and recourse

EAIF ADDS EXPLICITLY

- Reason · Access · Act execution architecture
- Architecture for workflows, agents, tools, and enterprise actions
- Operational boundary between platform mechanism and domain meaning

> **The mapping represents alignment of concerns, not equivalence or conformance.**

EAIF consumes applicable requirements from governance and risk-management processes. It does not replace them.
Source: NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0), Govern · Map · Measure · Manage.

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EAIF in Practice

Conversational Workforce Analytics

> **“Show me my headcount by location and explain significant changes from last quarter.”**

Business objective: provide a correct, authorized workforce insight with less effort. The objective is not to generate SQL.

Architecture area	Application
CONTROL	Identity: manager + authorized scope Purpose: headcount by location + quarter comparison Policy: authorized population, read-only, protected data
EXECUTE	Reason: interpret, analyze, explain Access: workforce semantics, hierarchy, current + historical data Act: not required
DOMAIN	Workforce owns headcount definition, authoritative sources, semantics, and business correctness
ASSURE	Protect: preserve scope and read-only behavior Evaluate: technical + business correctness Observe: sufficient execution and evaluation evidence
EVOLVE	Improve semantics, validation, clarification, execution mechanisms, or deterministic handling based on evidence

Key Lessons

- Not every intelligent capability needs Act.
- A technically valid query can still be a wrong business answer.
- The platform provides reusable mechanisms; the Workforce domain owns meaning and correctness.
- Improvement does not automatically mean adding more AI.

> **EAIF structures responsibility without prescribing the technology stack.**

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EAIF Quick Reference

Design

TIMELESS	SIMPLE	BORING	LEAN

Principles

Business Value First · Centralize Mechanisms, Decentralize Meaning · Trust Through Evidence

EAIF Core

Domain	Question	Capabilities
CONTROL	Should this happen, and under what conditions?	Identity · Purpose · Policy
EXECUTE	How is useful work performed?	Reason · Access · Act
ASSURE	Can we trust what happens?	Protect · Evaluate · Observe
EVOLVE	What evidence should make it better?	Improve

10 Architecture Questions

1. What business outcome should improve?	6. What authorized information and context are required?
2. Who initiated the work, who or what is acting, and under whose authority?	7. Does execution need to act? If so, through which enterprise capability or system?
3. What is the purpose?	8. What protection and evaluation are justified by the consequence of being wrong?
4. What policy, permissions, constraints, delegation boundaries, and approval requirements apply?	9. What must be observable to operate and account for material execution?
5. What reasoning is required?	10. What evidence should cause the capability or architecture to change?

> **Control what is permitted. Execute useful work. Assure what happens. Evolve from evidence.**

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